

16-March-24

Solomon Casspi - Recommendation Letter

To whom it may concern,

It is my great pleasure to write this letter in support of Solomon's PhD application. Solomon was a PhD student under my supervision for one year. We have recently parted in good terms, due to two main reasons: The most important one being Solomon's desire to seek expertise in control, an area in which I am not an expert. The other factor which I believe was important here was my absence and inability to properly supervise Solomon during the months following the October 7th attack, where Solomon was in a point where he needed my guidance and I was away for a lengthy reserved service. I would personally be happy to continue working with Solomon, yet I completely respect his decision to seek PhD studies elsewhere, and I believe his future PI would be very fortunate to have such a hardworking and dedicated student.

During our joint work, Solomon's focused on a core problem in data science: data-aided design of control system. His work dealt with the exploration of the emerging methodology of model-based deep learning in its unexplored domain of stochastic control, aiming to convert classical control algorithm into discriminative machine learning models. Doing so enabled Solomon to overcome fundamental challenges in stochastic control, such as the inherent limitations of separation-based designs, while harnessing principled mathematical models and classical algorithmic steps. Despite the fact that Solomon is has only completed a single year of PhD studies, he already managed to make substantial contributions, and his novel derivation of a data-aided linear quadratic Gaussian controller was accepted for publication in the leading conference of the signal processing society – IEEE International Conference on Acoustics, Speech, and Signal Processing.

Solomon's masters research dealt with compensation for mismatches and calibration in automotive radar. In his studies, Solomon combined principled mathematical concepts from estimation theory with hardware-aware and application oriented algorithmic derivations. His research work shows both high analytical skills, which are essential for successful theoretical studies, as well as dedication, hard work, creativity, and independence. Solomon's masters' results gained recognition as scientific paper was published in the prestigious IEEE Transactions on Aerospace and Electronic Systems, indicating his research achievements. In my opinion, the ability to integrate theoretical knowledge with concrete applicative and hardware considerations, as demonstrated in Solomon's research, is a key and non-trivial skill,

which is extremely advantageous in the research areas of control which he intends to pursue in his PhD.

As a final note, I believe that Solomon possesses all the abilities of an outstanding graduate student: he is sharp-minded, extremely hard-working and dedicated, and does not leave a stone unturned in his research work. I am certain that with the proper supervision, he can make ground-breaking contributions in the area of control and its integration with data sciences.

Feel free to contact me if you have any questions.

Sincerely,

Dr. Nir Shlezinger



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